



## Koneru Lakshmaiah Education Foundation

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### OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

#### Project Problem Statement Submission Form

##### Section / Team Details

Section No.: 4

Team No.: 16

Project Title: Kernel Monitor

##### Team Members

| Roll Number | Student Name    | Signature |
|-------------|-----------------|-----------|
| 2520030256  | Saahas Vaddi    |           |
| 2520030023  | Lohith Sanapala |           |
| 2520030427  | Rithwik Rapaka  |           |

##### 1. Abstract

Kernel Monitor is a lightweight, terminal-based system monitoring utility developed entirely in the C programming language for the Ubuntu/Linux operating system. The project addresses the practical need for real-time visibility into system resource usage without depending on any graphical dashboard, web interface, or external service. It obtains live system state directly from the Linux `/proc` filesystem, reading CPU utilization statistics from `/proc/stat`, memory usage details from `/proc/meminfo`, and per-process information from `/proc/[pid]/status`. This data is parsed, processed, and displayed directly in the Ubuntu terminal, with information refreshed continuously after a fixed time interval to reflect the current system state.

A key feature of the project is its threshold-based alerting logic, which compares real-time CPU and memory usage against predefined limits and displays an alert in the terminal whenever either threshold is exceeded, allowing early detection of abnormal resource consumption. All monitored data and alerts are additionally written to a log file, enabling users to maintain a historical record of system activity for later review. From a systems programming perspective, the project reinforces core Operating Systems concepts, including process management, memory management, CPU and resource monitoring, file handling, and real-time interaction with the Linux `/proc`

filesystem, all implemented through low-level, system-level programming in C. The expected outcome is a functional and resource-efficient command-line tool that enables users to observe CPU usage, memory usage, and running processes in real time, detect abnormal system behaviour through threshold-based alerts, and maintain accurate historical logs, all within a purely terminal-based environment.

## **2. Problem Statement**

Diagnosing performance issues on Linux systems currently requires switching between multiple native tools (top, ps, vmstat, iostat), each showing only a narrow slice of system state with no unified view, historical logging, or automated alerting. This makes it difficult for developers, students, and system administrators to quickly identify the root cause of high CPU load, memory pressure, or a runaway process, and to track how resource usage evolves over time. The problem is directly relevant to Operating Systems and Systems Programming, since solving it requires understanding process scheduling, memory management, and how the kernel exposes system state through interfaces such as /proc and system calls, and translating that low-level data into an accessible, real-time view for the end user.

## **3. Objectives**

1. Design a Linux-based data collection module that reads process and system metrics (CPU, memory, disk I/O) from /proc and relevant system calls.
2. Implement a lightweight backend service to aggregate, store, and expose collected metrics through an API.
3. Build a real-time environment to visualise live resource usage, historical trends, and per-process breakdowns.
4. Implement configurable threshold-based alerting to notify users of abnormal CPU, memory, or process behaviour.

## **4. Proposed Methodology**

The project will be developed and tested on a Linux (Ubuntu) environment. A C data-collection module will periodically read system and process information from the /proc filesystem (e.g., /proc/stat, /proc/meminfo, /proc/[pid]/status) and invoke relevant system calls (getrusage, sysinfo) to gather CPU, memory, and I/O statistics. Collected data will be serialized and sent to a backend service (Python/Node.js) over a local socket or REST API, where it is validated, stored in a database.

A configurable rules engine will compare incoming metrics against user-defined thresholds and trigger notifications when limits are exceeded. The system will be containerised using Docker for ease of deployment and testing across environments.

## 5. Operating Systems Concepts / Linux APIs Used

| OS Concept / Linux API / System Call                             | Purpose in the Project                                       |
|------------------------------------------------------------------|--------------------------------------------------------------|
| /proc filesystem (/proc/stat, /proc/meminfo, /proc/[pid]/status) | Read live CPU, memory, and per-process statistics            |
| getrusage()                                                      | Retrieve resource usage information for processes            |
| sysinfo()                                                        | Obtain overall system memory and load statistics             |
| Process & Thread Management                                      | Track active processes/threads and their states              |
| POSIX Sockets / IPC                                              | Transfer collected metrics from collector to backend service |

## 6. Individual Contribution

| Roll Number | Student Name    | Individual Responsibility                      |
|-------------|-----------------|------------------------------------------------|
| 2520030023  | Lohith Sanapala | Kernel-level data collection module            |
| 2520030256  | Saahas Vaddi    | Backend service, API, and database integration |
| 2520030427  | Rithwik Rapaka  | visualisation, and alerting logic              |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                                                   |
|----------------------------|-----------------------------------------------------------|
| Ubuntu                     | Development and testing OS environment                    |
| C                          | Kernel-level data collection using system calls and /proc |
| Python (FastAPI) / Node.js | Backend service for aggregation and API                   |
| MySQL / PostgreSQL         | Storage of historical metrics and logs                    |
| Git & GitHub               | Version control                                           |

## 8. Expected Outcome

The completed project will deliver a working Linux-based Kernel Monitor capable of collecting and displaying real-time CPU, memory, and process-level statistics through an interactive dashboard. It will maintain historical logs of system resource usage, support configurable alert thresholds, and notify users when abnormal behaviour is detected. The tool is expected to give developers, students, and administrators a clearer, consolidated view of kernel-level activity than existing fragmented command-line utilities, aiding faster diagnosis of performance issues.

## 9. GitHub Repository

GitHub Repository Link: [Loitcraft-65/KLH\\_CSE\\_2026-2027\\_TEAM-16\\_OSSP\\_Kernel\\_Monitor](#)

Repository Created: \_\_\_\_\_

☐ Yes ☐ No

### Faculty Name and Signature

Faculty Name: \_\_\_\_\_

Signature: \_\_\_\_\_

Remarks: \_\_\_\_\_

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### Faculty Use

Project Approved: \_\_\_\_\_

☐ Yes ☐ No

Date: \_\_\_\_\_

Faculty Signature: \_\_\_\_\_